

Sophia Li

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PROFILE

Mechanical Engineering student at Stanford who takes hardware projects from first-principles analysis through CAD, fabrication, and physical test, independently and under tight constraints. Skilled in Fusion 360, SolidWorks, Abaqus, and MATLAB, with a track record of owning a project end to end and rebuilding understanding from fundamentals when test results don't match predictions.

EDUCATION

Mechanical Engineering, Stanford University

Palo Alto, CA | **Sept. 2025 – Jun. 2029**

GPA: 4.048

- Relevant Coursework: AA151 – Aerospace Structures, AA126 – Aerospace Prototyping Lab, ENGR14 – Statics, ME30 – Thermodynamics, CS223A/ME320 – Introduction to Robotics (Graduate), AA100 – Introduction to Aeronautics and Astronautics

SKILLS

- **CAD & FEA:** Fusion 360, SolidWorks, Abaqus, GD&T (ASME Y14.5), HyperX
- **Structural Analysis:** Classical Laminate Theory, Composite Structures, Mechanics of Materials, Failure Criteria
- **Hardware & Manufacturing:** Flight Hardware Development, Rapid Prototyping, Design for Manufacturing (DFM), Composites Manufacturing (wet layup), 3D Printing, Fabrication Tooling, Soldering, Circuit Wiring, Woodworking
- **Aerodynamics:** XFLR5, Airfoil Analysis, Lift Distribution Modeling, Wing Sizing
- **Programming:** Python, C++, MATLAB, Microsoft Excel, Google Suite (Gmail, Sheets, Docs)

ENGINEERING PROJECTS

Composite Wing Design, Analysis, Manufacturing, & Testing

Jun. 2026 – Jul. 2026

- Designed a carbon fiber composite foam-core NACA 2412 wing panel in Fusion 360, generating CAD models and engineering drawings to carry the full structural design cycle from load definition through physical prototype and test.
- Fabricated flight-representative composite wing hardware using wet layup and foam-core manufacturing techniques, selecting spar and skin materials based on stiffness and thermal service limits, and built 2D and 3D Abaqus composite shell models that predicted structural response within 2% of analytical calculations.
- Performed hand calculations and numerical finite element analysis, using HyperX Tsai-Wu and buckling failure theories, to evaluate structural design margins for both strength and damage tolerance in a lightweight, load-bearing composite structure, while also accounting for thermal service limits in material selection.
- Developed MATLAB scripts to calculate laminate stiffness, neutral axis location, and bending response from first principles beam theory, informing layup and spar design decisions.
- Designed, built, and ran a whiffletree tester to validate a composite wing panel under full design load, collecting data and performing post-test analysis; when measured deflection exceeded prediction, root-caused the discrepancy to manufacturing-driven stiffness loss and defined a design change for the next build.

Custom Hot Wire Cutter Design & Fabrication

Jul. 2026

- Independently developed a novel, low-cost hot wire cutter to solve a fabrication problem, engineering the frame for rigidity and precise wire tensioning using hand tools and basic shop equipment to prevent warping or drift mid-cut, supporting fabrication of the composite wing above.
- Performed thermal analysis to calibrate a 12V power supply (repurposed from an old PC), evaluating current draw and resistance tradeoffs to achieve consistent cutting temperature without risking burnout.
- Built the tool from largely repurposed and low-cost materials (leftover wood panels, ~\$10 in new components), reflecting a creative, resourceful, problem-first approach to fabrication tooling.

EXPERIENCE

Mechanical Engineering Composites Intern, Earthly Dynamics Corporation

Atlanta, GA | **Mar. 2027 (Incoming)**

- Selected to support composite structures design, analysis, and fabrication for unmanned aerial systems.

Mechanical Design and Structures Lead, Stanford DBF

Stanford University | **Sept. 2025 – Present**

- Owns structural sizing for the full aircraft, including wing spar, fuselage, empennage, and landing gear, translating aerodynamic and mission loads into sized structural members while coordinating structural and design decisions across aerodynamics and manufacturing teams.
- Leads mechanical design and prototyping for an unmanned aircraft competing in the annual AIAA Design/Build/Fly competition, coordinating cross-functionally across a 10-engineer team through design reviews from initial design through fabrication and test.
- Designed, built, and tested a two-stage servo-actuated mechanism to release and drop a banner mid-flight, integrating an actuator and servos into a compact assembly to meet flight hardware constraints.
- Worked directly with team members on Fusion 360 CAD models and documentation, maintaining manufacturing documentation and drawings.

Research Assistant, Stanford Freeman Spogli Institute (FSI)

Stanford University | **Apr. 2026 – Present**

- Sole author of the lab's IRB application, applying rigorous technical and academic writing to meet ethical and safety compliance standards for human-subjects testing.
- Executed quantitative data analysis of large-scale survey data on care robot technology adoption using Microsoft Excel, developing visualizations to communicate findings to faculty collaborators.

Outreach Officer, Stanford STEM Nights

Stanford University | **Oct. 2025 – Present**

- Coordinated cross-functional teams to deliver technical demonstrations for large-scale STEM community service events, ensuring safety compliance and effective public communication.

AWARDS

- 2024 + 2025 Morley Award Recipient (~23/1161) — awarded to Georgia Tech's distance math students whose final grades in linear algebra and multivariable calculus ranked in the top 2% of the class.